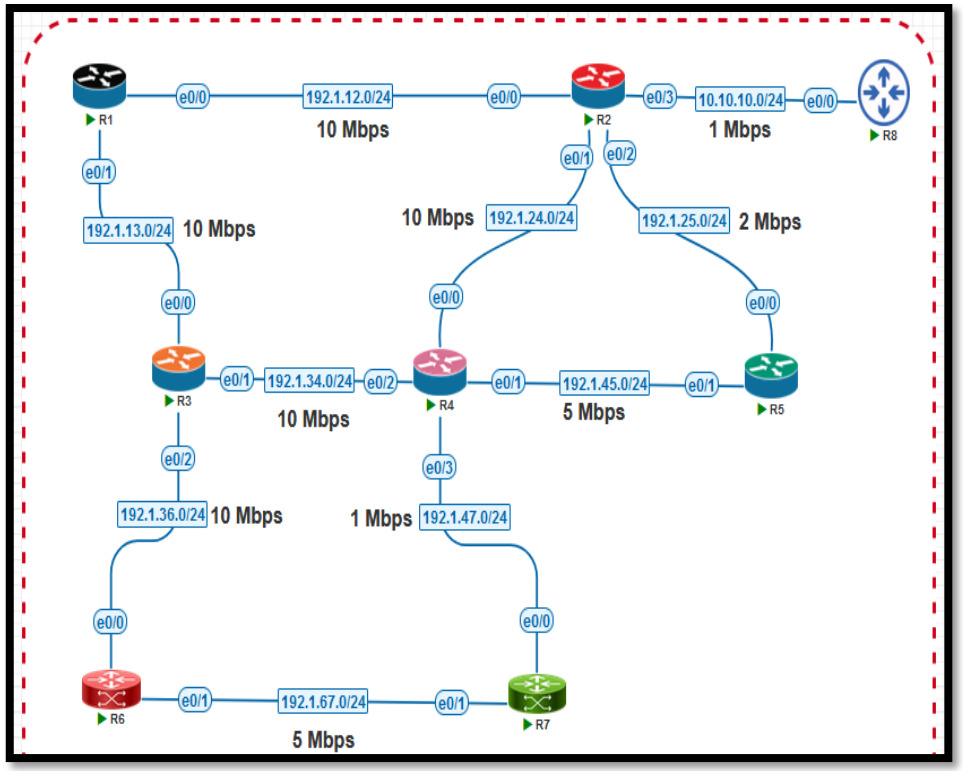
Lab 1 – Initializing EIGRP – Network Statement



Task 1 – Configure R1 in EIGRP AS 111.

- Configure the Interface based on the Interface Configuration Table.
- Configure the Network statement **based on Major networks only.**

Task 2 – Configure R2 in EIGRP AS 111.

- Configure the Interface based on the Interface Configuration Table.
- Configure the Network statement using a **wild card mask for the Network 10.0.0.0**
- Use the Major networks for the rest of the network

Task 3 – Configure R3 in EIGRP AS 111.

- Configure the Interface based on the Interface Configuration Table.
- Configure **one Network statements** to enable EIGRP on the networks starting **with 203.X.X.0**
- Use the Major networks for the rest of the networks.

Task 4 – Configure R4 in EIGRP AS 111.

- Configure the Interface based on the Interface Configuration Table.
- using a **wild card mask to minimize** the Network statements **starting with 104.X.X.0.**
- Use the Major networks for the rest of the networks.

Task 5 – Configure R5 in EIGRP AS 111.

- Configure the Interface based on the Interface Configuration Table.
- using a **wild card mask to minimize** the Network statements **starting with 205.X.X.0**
- Major networks for the rest of the networks.

Task 6-7 – Configure R6-R7 in EIGRP AS 111.

- Configure the Interface based on the Interface Configuration Table.
- using a **wild card mask to minimize** the Network statements **starting with 101.X.X.0**
- Major networks for the rest of the networks.

Task 8 – Configure R8 in EIGRP AS 111.

- Configure the Interface based on the Interface Configuration Table.
- Configure the Network statement **such that all directly connected interfaces are enabled in EIGRP**. This should also take care of **any new interfaces configured run EIGRP in the future**.

==========

Lab 2 – EIGRP – Passive Interfaces

Task 1 – Configure Passive-Interface on Routers in EIGRP AS 111.

- Configure R1 such that do not send updates on **links that do not have other routers**, basically Loopbacks.
- Use the minimum number of passive interface commands to accomplish the task.

Lab 3 – EIGRP – Unicast Neighbors

Task 1 – Configure Unicast EIGRP on specific interfaces in AS 111.

- Configure the neighbor relationship between R1 and R2 to be Unicast-based.
- Configure the neighbor relationship between R2 and R8 to be Unicast-based.

==========

Lab 4 – EIGRP – Metric Calculations

256*[10^7/BW/(min)+Delay(sum)/10]

Task 1 – Configure interface bandwidth based on the topology diagram.

- Configure the Interfaces on the routers based on the bandwidth shown in the diagram.
- Calculate the metrics from R2 towards the 205.1.4.0/24 network Make sure that it matches the values

Task 2 – Configuring Unequal Load Balancing on R2

- Configure R2 to **use both path towards the Loopback Interfaces of R5**.
- Calculate the Variance and implement it on R2.

Note

Return route from R5 will be R4 Cause it's the best Route

Task 3

Configure all EIGRP routers **so that only delay is used** in the composite **metric calculation.**

Lab 5 – Route Summarization – Auto Summary

Task 3 – 1st check auto-Summary on Routers & Configuring Auto-Summary on R2 & R8

Task 4 – Verifying Auto-Summarization between R2 & R8

- Are the Network 10.0.0.0/8 subnets summarized between R2 & R8?

Same major Network (contiguous)

Task 5 – Configuring the following Interfaces on R2 & R8

Enable them in EIGRP.

```
R2
Interface loopback401
Ip address 102.1.4.1 255.255.255.0
Interface loopback402
Ip address 102.1.5.1 255.255.255.0
Interface loopback403
Ip address 102.1.6.1 255.255.255.0
Interface loopback404
Ip address 102.1.7.1 255.255.255.0
!
Router Eigrp 111
Network 102.1.0.0   0.0.255.255
```

```
R8
Interface loopback401
Ip address 108.1.4.1 255.255.255.0
Interface loopback402
Ip address 108.1.5.1 255.255.255.0
Interface loopback403
Ip address 108.1.6.1 255.255.255.0
Interface loopback404
Ip address 108.1.7.1 255.255.255.0
!
Router Eigrp 111
Network 108.0.0.0
```

Task 6- Verifying Auto-Summarization between R2 & R8

- Are the new Loopback Networks summarized between **R2 & R8?**
- Why

Different major Network(Discontiguous)

Lab 6 – Route Summarization – Manual Summarization

Disable Auto-Summarization on R2& R8

Task 1 – Configuring Manual Summarization

- on R1 for all the 101.0.0.0/8 subnets using the Longest Summary Mask.
- Summarization should be configured towards all the neighbors. (configure All Routers to practice)

==========

Lab 7 – Route Summarization – Leak Maps

Task 1 – Configure manual summarization R7 such that

- Traffic for the 101.1.75.0/24 network from R3 should use R4 as the Next-Hop
- whereas the other networks should continue to use the Summary Route based on the best metric.

Lab 8– Authenticating EIGRP Neighbors using MD5

Task 1

Configure MD5 authentication for all links between R2& R8 & **key Rotation**.

==========

Lab 9 – Configuring a Basic Named Mode Configuration

Task 1 – Re-Configure R1, R2, Using Named-Mode EIGRP & Same AS number

-Network statement + Wildcard Mask
-Passive interface & default
-Unicast Neighbor
-Unequal Load Balance
-Manual Summarization
-Leak Map
-Authentication md5 &SHA

EIGRP Filtering using ACL
Task

Configure **R1** to deny **8.8.8.8/32**

EIGRP Filtering using ACL
Task

(EVEN networks) 2-4-6 R8

* Create loopbacks **on R8**
* Configure **R2** block All **ODD** networks in **172.16.X.0** range (1-3-5)

Loopback 21: 172.16.1.1/24
Loopback 22: 172.16.2.1/24
Loopback 23: 172.16.3.1/24
Loopback 24: 172.16.4.1/24
Loopback 25: 172.16.5.1/24
Loopback 26: 172.16.6.1/24

EIGRP Filtering using ACL
Task

* Create loopbacks **on R1**
* Configure **R2** block these specific networks : One ACL

-150.55.74.0/24
-150.55.90.0/24
-150.55.202.0/24
-150.55.218.0/24

Loopback200	150.55.50.1
Loopback201	150.55.74.1
Loopback202	150.55.80.1
Loopback203	150.55.90.1
Loopback204	150.55.150.1
Loopback205	150.55.175.1
Loopback206	150.55.200.1
Loopback207	150.55.202.1
Loopback208	150.55.210.1
Loopback209	150.55.218.1

EIGRP Filtering with Prefix-Lists

Task

configure **R8** to stop advertising network **10.1.11.0/24 to R2 .**

Configure the following Loopback Interfaces **on R5**

Advertise them under EIGRP 111:

Loopback 401: 175.50.1.1/24
Loopback 402: 175.50.2.1/24
Loopback 403: 175.50.3.1/24
Loopback 404: 105.1.1.33/27
Loopback 405: 105.1.1.65/28

EIGRP Filtering with Prefix-Lists

Task

configure **R2& R4** to receive any updates from **R5** with a prefix-length **of 8 to 24**. Do not configure the filtering on R5.

Configure the following Loopback Interfaces on R6 advertise them under EIGRP 111:

Loopback 401: 176.1.32.1/19
Loopback 402: 176.1.64.1/20
Loopback 403: 176.1.80.1/21
Loopback 403: 176.1.96.1/24

EIGRP Filtering with Prefix-Lists

Task

configure **R6** such it does not send networks **176.1.0.0/16 major network** range that have a mask **greater than 20**.

EIGRP Offset List

Task

Configure an **offset-list on R4** so that **R7** traffic destined **for R4's Loopback0** prefix is routed **through R6.**

EIGRP Filtering with Per-Neighbor AD

Task

Configure administrative distance **filtering on R7** so that traffic destined **for R4's Loopback0** prefix is routed **through R6.**

EIGRP Filtering with Administrative Distance

Task

Configure administrative distance filtering **on R2** so that devices within the network **cannot reach R8's Loopback0**

EIGRP control traffic with summary

Task

Configure EIGRP summarization on **R2** for networks **202.1.4.0/24 202.1.5.0/24 202.1.6.0/24 202.1.7.0/24** **so that R5** reach these networks **through R4**

Task – **(Failure Detection)**

Configure BFD [Bidirectional forward detection]

Configure routers in EIGRP in AS 111
Such that all neighbor down events are detected **in a Sub-second** times to optimize reconvergence of a network

***** Use a BFD **Hello Interval of 50 msec**.
***** The dead time should be **3 times the Hello Interval**.

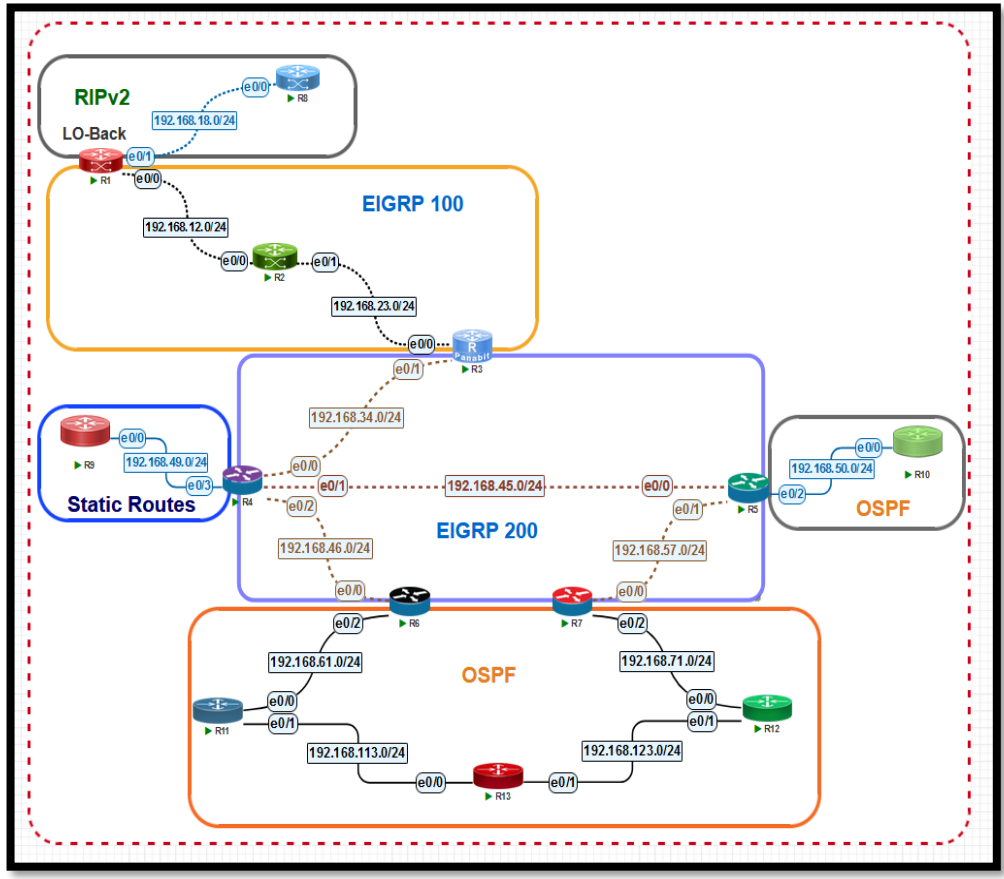
Task 1

You would like to provide reachability between the **Static Routing** and **EIGRP 200 domains**
R9 is already configured with a **Default Route towards R4.**

Verification:

- Try Pinging 150.1.9.9 (R9's Loopback) from R5. Are you successful?
- Try Pinging 192.168.49.9 (R9's E0/0 IP) from R5. Are you successful?
- Try Pinging 192.168.45.5 (R5's E0/0 IP) from R9. Are you successful?
- The reason is that 192.1.49.0/24 is a **directly connected interface on R4**.

❑ **Redistribute Static does not include the Connected Route.**



Task 2

- Redistribute the connected route into EIGRP.

Make sure to only **redistribute the R4-R9 directly connected Interface**.

Task 3

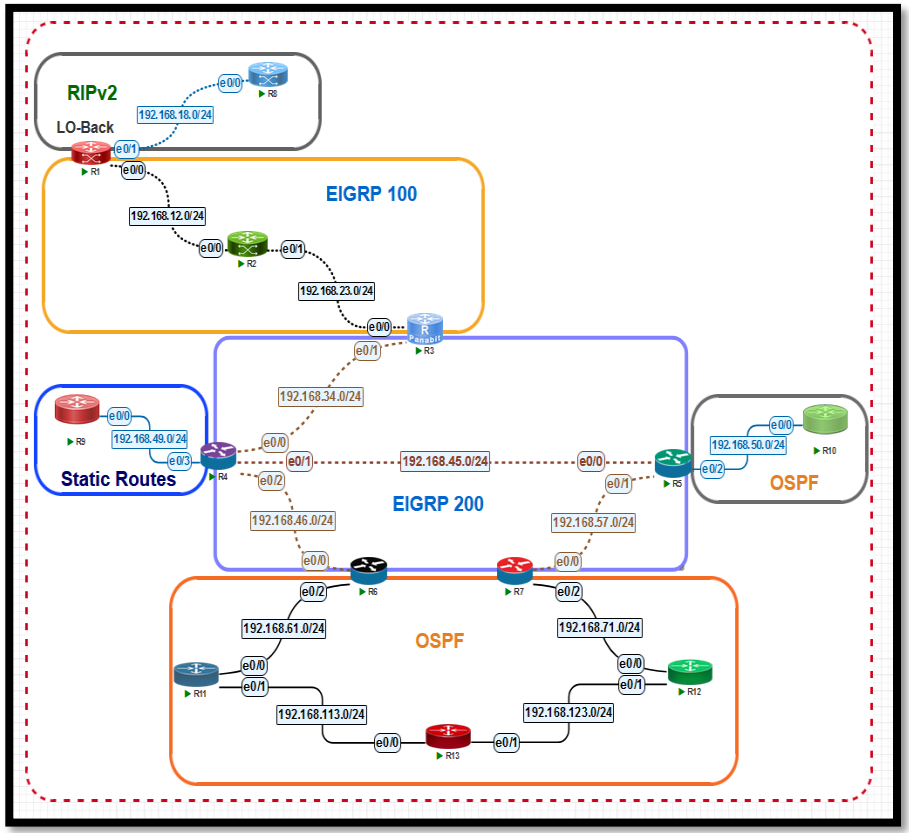
Networks **202.1.1.0/24 & 202.1.3.0/24** networks should not be **Redistributed into EIGRP 200**.
Re-configure Redistribution **on R3** to accomplish the requirement.

Route-map With ACL

Task 4

Networks 9.9.9.9/32 network should not be **redistributed into EIGRP 200**
Re-configure Redistribution **on R4** to accomplish the requirement.

Route-map With prefix-list



Redistribution with Route Tagging

Task 5

You are required to **block the RIPv2 routes** from propagating **into the OSPF domain and vice-versa**.
Use a Mechanism that ensures that if **new routes are added** into the RIP or OSPF domains
they **continue to get blocked** from propagating to each other **without having** to do any **extra configurations**

Multi-Point Redistribution with Route Tagging

Task 6

EIGRP 200 & OSPF need to redundancy with Redistribution, **needs to be configured on R6 & R7.**
Make sure that the **routes are blocked** from coming back **into the Source Domain**.